

Mars image anomaly detection

NSSC 2026 | Measured experiments and final analysis

I trained an autoencoder from scratch and used Isolation Forest on its latent vector. The first candidate list was dominated by unusually bright images and changed across repeated runs. I added controlled brightness and footprint experiments to investigate that weakness.

My decision

I retain v3 as the screening reference. v7 improves validation MSE by 1.68% and gradient error by 0.52%, but four of its five selected crops remain extremely bright and forest-repeat agreement weakens. v4-v6 reduce specific nuisance responses while introducing border or fill concerns; v6 fails its prespecified promotion criteria. v8 has zero image-only flags. No tested replacement establishes better hidden-anomaly detection. v3 itself remains brightness-confounded and unstable across independent runs; its flags are investigation candidates, not confirmed anomalies.

Reference run: v3. The calibrated boundary is 0.542471; 17 of 10,422 crops exceed it. The 95% source-bootstrap threshold interval is [0.535230, 0.550470].

What this establishes. The experiments demonstrate modest reconstruction gains and controlled nuisance invariance. They do not establish a detection-accuracy improvement. I retain failed ablations and report negative outcomes.

What it does not establish. Hidden anomaly labels are unavailable. I cannot claim precision, recall, F1 or a verified false-positive rate. Candidate appearance and reproducibility are not ground truth.

Run	Input	Val. MSE	Val. SSIM	Flags
v1	raw	0.002694	0.6611	20
v2	raw	0.002740	0.6717	9
v3	raw	0.002599	0.6738	17
v7	raw	0.002555	0.6746	6
v8	raw	0.002678	0.6686	0

Metrics use each run's best validation-objective checkpoint. Different preprocessing changes the reconstruction target, and the split repeat uses different validation images. Their MSE/SSIM are not interchangeable quality scores. All runs use random initialization.

The notebook is prepared for Colab/Kaggle. These saved experiments ran on a local RTX 3050; no cloud execution is invented.

Data and the preprocessing comparison

I decoded all 10,422 supplied crops, verified 227 by 227 grayscale inputs, checked metadata joins and found no exact decoded-pixel duplicates. There are 172 source observations. Near-duplicate or geographically overlapping observations remain possible.

Main split	Sources	Crops
Training	120	7,101
Validation	25	1,666
Calibration	27	1,655

I keep each source in one split. v4 changes only preprocessing: subtract the crop mean, divide by standard deviation, map with $0.5 + 0.15 \cdot z$, then clip to $[0,1]$. Standard deviation is floored at $1/255$. v5 adds exclusion of exact-zero pixels connected to the image border when estimating the transform, then fills them with 0.5. Disconnected interior zeros remain content.

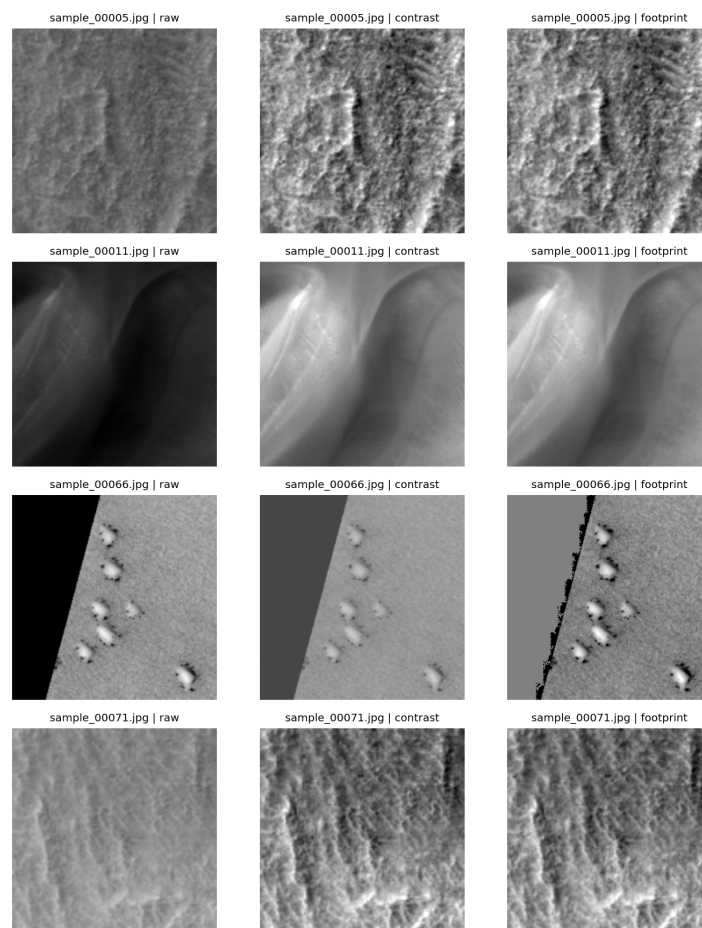


Figure 1. Fixed validation examples: raw, contrast-standardized and footprint-treated inputs. The mask is a heuristic, not an organizer-supplied validity mask.

This intentionally suppresses global brightness and can remove meaningful albedo information. Clipping can suppress extreme contrast; normalization can amplify noise; a connected shadow can be mistaken for an image footprint.

Phase 1 Architecture, loss and learning curves

Five stride-two convolution blocks use channels 16, 32, 64, 96 and 128, with GroupNorm and SiLU. Spatial sizes are 114, 57, 29, 15 and 8. A linear layer creates a fixed 128- or 256-dimensional vector. The decoder expands the vector and uses resize-convolution to reconstruct exactly 227 by 227. There is no bypass connection and no pretrained feature extractor.

v1 uses MSE. v2-v6 use $0.9 \text{ MSE} + 0.1 (1 - \text{SSIM})$, with an 11 by 11 Gaussian SSIM window, sigma 1.5 and data range 1. AdamW uses learning rate 0.0003, weight decay 0.0001 and batch size 16. Each run completed 15 epochs, with best validation checkpoint selection. v7 adds 0.1 times adjacent-pixel gradient error; v8 retains that loss and adds batch flips and quarter-turn rotations. The actual MSE coefficient remains 0.9. The original v7 protocol described 0.8; that deviation is recorded rather than hidden.

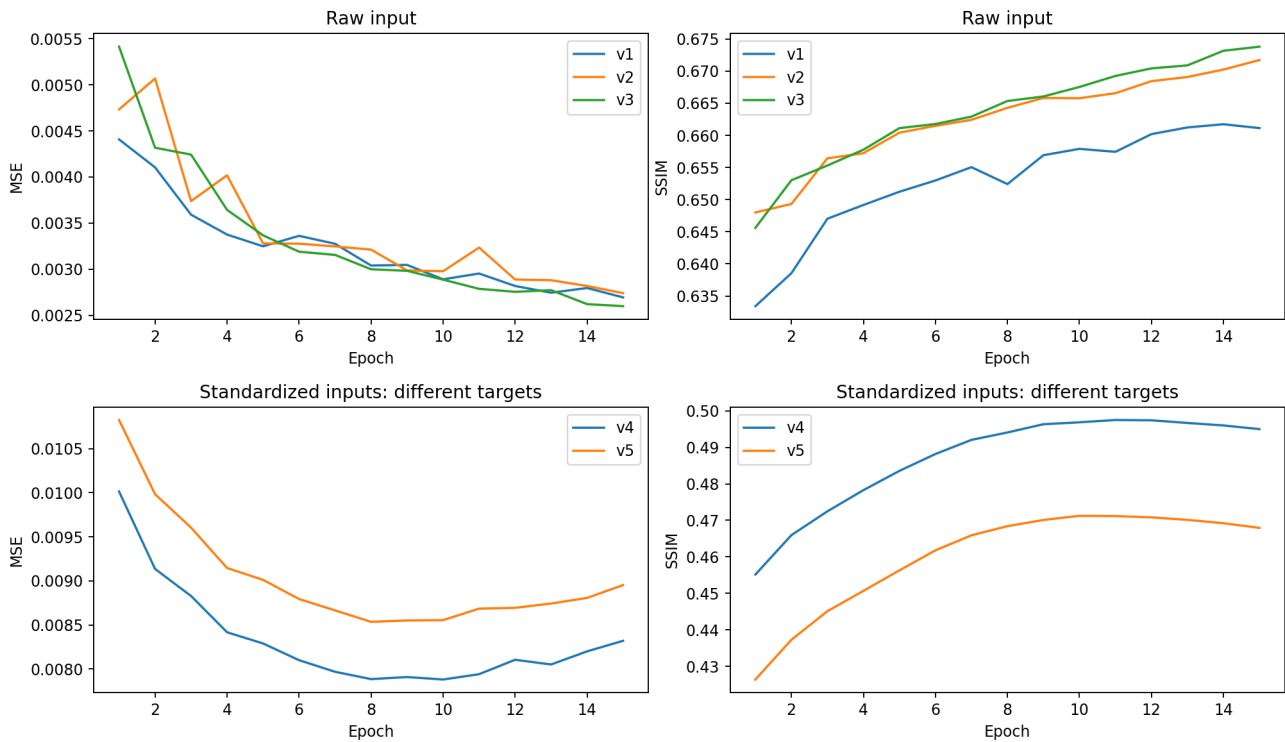


Figure 2. Raw-input and standardized-input learning curves are separated because their reconstruction targets differ.

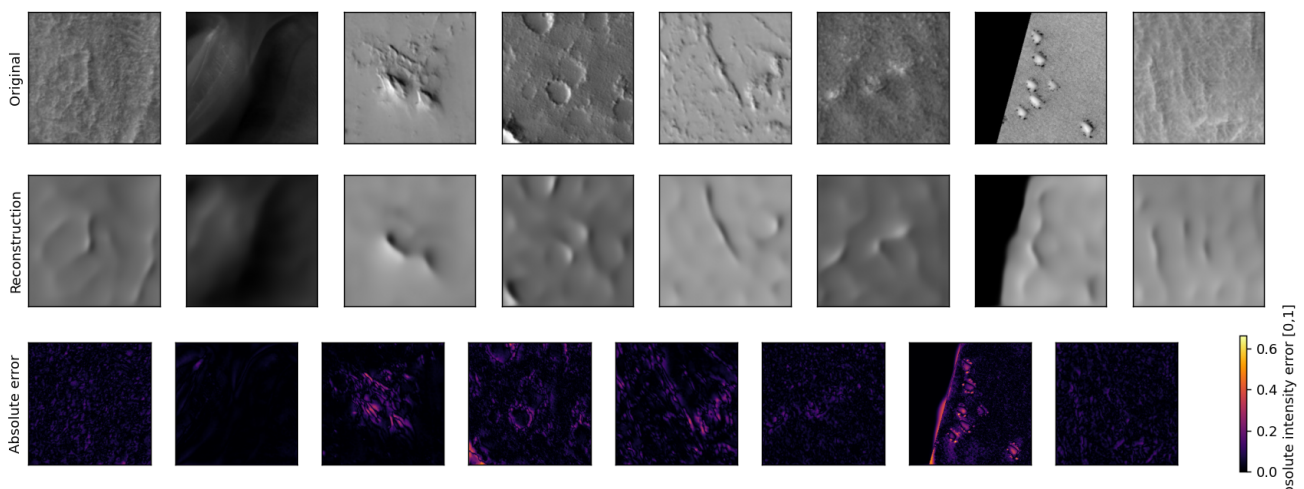


Figure 3. Fixed model inputs, latest reconstructions and error. Scoring uses the best checkpoint. The reference reconstructs raw inputs; standardized ablations label their different input domain explicitly.

Phase 2 Scoring and statistical calibration

I fit 2,000 Isolation Forest trees, max_samples 256, on training latents only. Novelty is negative score_samples: higher means more unusual. Metadata and reconstruction errors are not forest features, and the library prediction offset does not determine the flags.

The original adjusted-boxplot threshold for v1 was 0.87484, above its maximum score 0.55648. Its multimodal histogram motivated a distribution-model revision; I retained the zero-flag result. I did not lower a threshold merely to produce five images.

For every run, I fit one through four Gaussian mixture components on calibration scores, select minimum BIC, and define $T = \max_j(\mu_j + 3 \sigma_j)$. All fitted populations contribute to the envelope. No assumed anomaly count or contamination fraction determines the result.

The selected run uses 4 components, with fitted tail mass 0.000306. This is a fitted-model description, not a real false-alarm guarantee. Correlated crops make BIC heuristic, and unknown contamination can distort the model.

The component count reaches the search limit of four, which is a model-selection limitation.

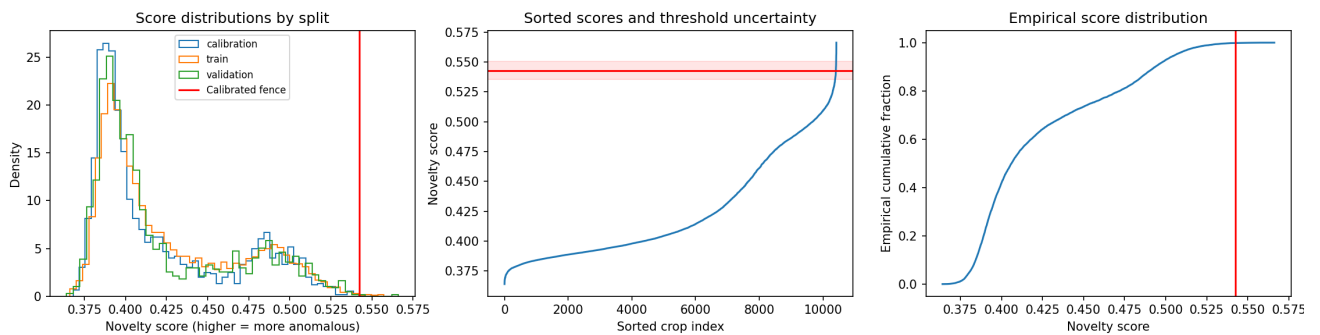


Figure 4. Split score distributions, sorted novelty scores and the calibrated threshold with source-bootstrap uncertainty.

I resample whole calibration sources and refit component selection in 100 bootstrap repetitions. The validation CDF maximum distance is 0.0597; I treat it descriptively and do not report an IID KS p-value. Conditional threshold flag frequency is not a probability of a genuine anomaly.

Did the improvement survive the checks?

For v4, changing initialization gives flag-set Jaccard 0.207 and full-ranking correlation 0.867; changing the source split gives 0.360 and 0.866. The three contrast runs flag 24, 11 and 10 crops, with 4 in their intersection. On 1,253 crops held out from both source partitions, rank correlation is 0.897. For raw v3, the corresponding initialization/split Jaccards remain 0.150/0.294, with zero crops flagged in all three raw runs. These are limited sensitivity checks, not detection accuracy. Initialization repeats change both encoder and forest seeds; separate forest-only repeats isolate detector randomness.

Comparison	Rank (all crops)	Jaccard (all crops)
v3 / robust_seed	0.949	0.150
v3 / robust_split	0.957	0.294
v4 / improved_seed	0.867	0.207
v4 / improved_split	0.866	0.360
v7 / v7_robust_seed	0.961	0.067
v7 / v7_robust_split	0.963	0.294

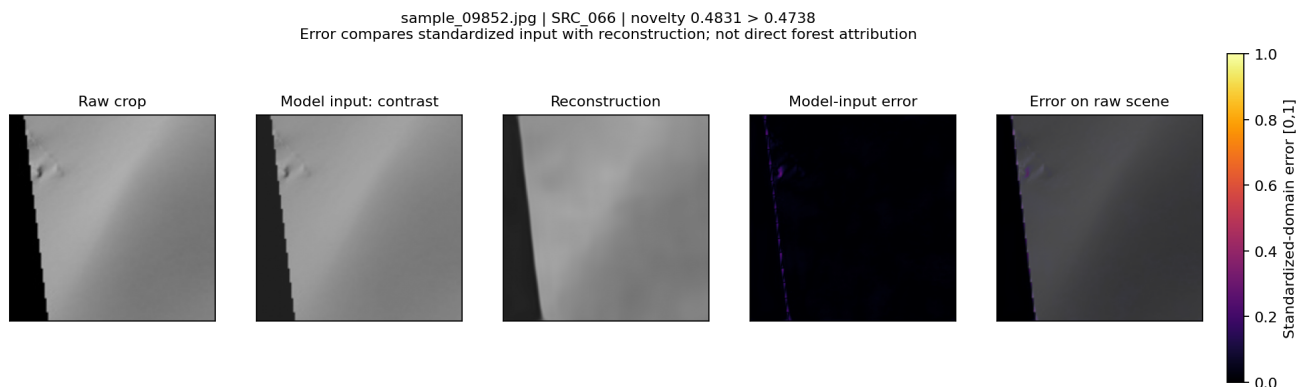
Jaccard is intersection divided by union of flagged sets. It can be low even when global ranking correlation is high. For v7 versus the split repeat, all-crop Jaccard is 0.294; it is 0 on the 1,253 common-held-out crops (zero versus one flag). The CSV labels both populations and denominators. Neither metric measures anomaly accuracy.

Run	Median score change	95th percentile change	Flag flips / 512
v3	0.007651	0.033935	0
v4	0.000000	0.002902	0
v5	0.000000	0.000044	0

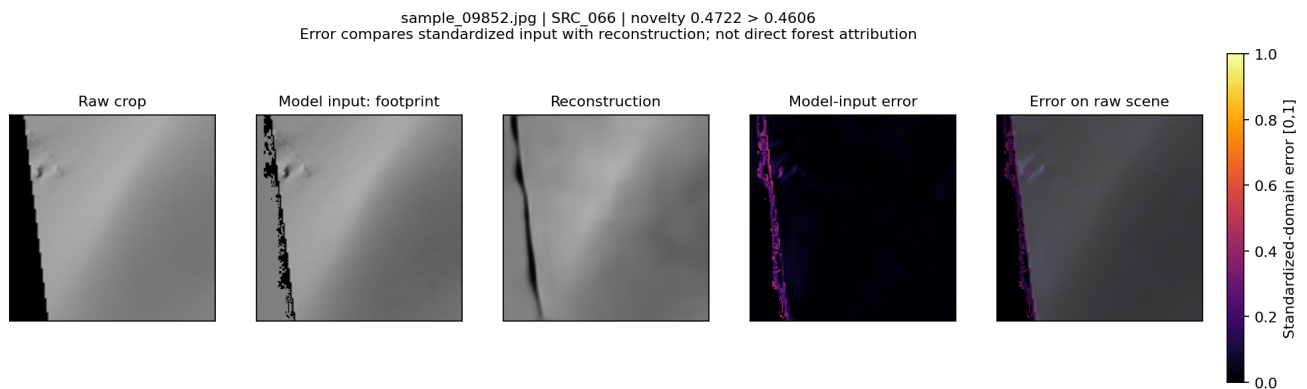
I hold each model, forest and threshold fixed, then apply $0.8 \cdot x + 0.1$ outside its original inferred black border on a fixed validation sample. This tests a specific photometric nuisance while preserving the footprint, not correctness on labelled anomalies.

Why better brightness invariance is insufficient

All five leading v4 candidates have conspicuous black strips or borders. Their brightness distribution is less extreme, but acquisition support remains a competing explanation. These examples were inspected after statistical selection; they were not used to lower the boundary or select a desired count.



v4: the same validation crop after the independently calibrated selection. The black border remains in the contrast-standardized input. Error belongs to the standardized input domain.

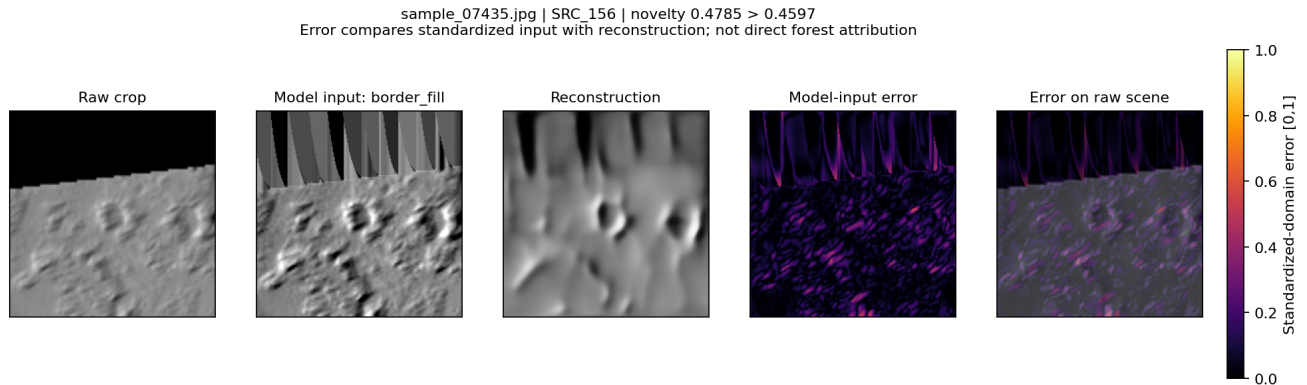


v5: the same validation crop after the independently calibrated selection. The footprint treatment leaves near-black edge fragments. Error belongs to the standardized input domain.

The controlled perturbation favors the invariance designed into normalization. It establishes that mechanism, but cannot establish preservation of real albedo anomalies. I retain v3 as the screening reference and preserve both negative ablations for the engineering journal.

Later experiments: v6, v7 and v8

v6 uses a fixed border-connected near-black mask (at most 4/255), two-pixel expansion and nearest-valid standardized fill. Its 95th-percentile border response is 0.002578 versus 0.030759 for v3, but the fraction of injected bright-square, dark-square and checker cases that increase novelty is only 30.1%, 60.9% and 32.4%. It fails the prespecified 70% requirement and the forest-stability requirement. No sampled perturbation crossed a threshold for any tested model.



The fill creates conspicuous stretched stripes. Across all crops, 2,577 have some masked pixels, three exceed 50% masking, and none are fully masked. I do not promote v6. Synthetic response checks are not hidden-label accuracy.

v7 improves validation MSE by 1.68%, SSIM by 0.000773 and gradient error by 0.52% relative to v3. Yet its top-five median brightness percentile is 99.952, four selections remain extremely bright, and minimum forest-repeat Jaccard drops from 0.667 to 0.417. The darker selection also contains black borders. This does not establish a solved brightness confound.

v8 adds geometric augmentation to v7. Its best raw-input validation MSE is 0.002678 and SSIM is 0.668622, worse than v7 on these measures. It has zero image-only flags. A reduced effective rank is descriptive, not proof of more meaningful features.

Phase 2.4 Optional metadata fusion

I compare matched five-forest ensembles on the same v8 image latents. Both use seeds 2026 through 2030, 2,000 trees per forest and 256 samples per tree. I average negative score_samples and fit each ensemble its own mixture boundary on calibration sources. Training-only scaling and one-hot encoding add sun angle, resolution and season before the forest, not the decoder.

Representation	Dimensions	Threshold	Flags
image_only	256	0.555633	0
fused	262	0.554207	0

All-crop score Spearman correlation is 0.997742. Added flagged filenames: []; removed filenames: []. Flag-set Jaccard is None (None means an empty union, not perfect agreement).

Zero thresholded candidates remain zero; a top-five list is not manufactured. The experiment measures the effect of metadata under this controlled comparison, not detection accuracy or a guaranteed bonus score. The original single-image-only versus fused-ensemble comparison is historical and cannot isolate metadata alone.

Both fitted ensembles, the fused scaler/encoder, score tables, input fingerprints and separate calibration records are saved. The required image-only v3 pipeline remains the primary result.

All experiment coverage

Run	Domain	Best epoch	Flags
v1	raw	15	20
v2	raw	15	9
v3	raw	15	17
robust_seed	raw	15	6
robust_split	raw	14	5
v4	contrast	11	24
v5	footprint	10	42
improved_seed	contrast	12	11
improved_split	contrast	11	10
v6	border_fill	10	78
v7	raw	15	6
v7_robust_seed	raw	15	10
v7_robust_split	raw	15	16
v8	raw	14	0

Phase 1.3 Latent-space diagnostics

The reference vector has covariance effective rank 31.53 out of 256. It summarizes the variance spectrum, not a number of terrain classes.

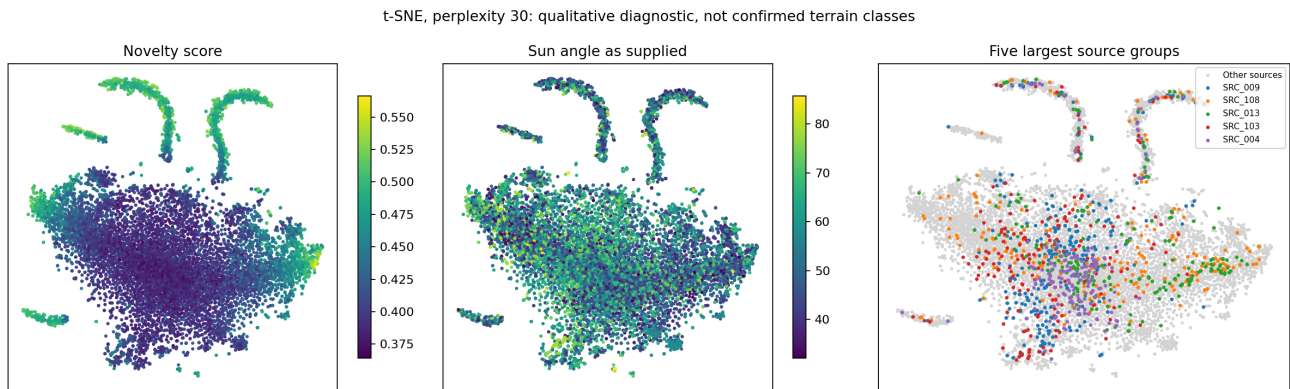


Figure 5. t-SNE at perplexity 30; novelty, supplied sun angle and five large source groups. PCA and standardization are for plotting only. The forest uses the full original latent.

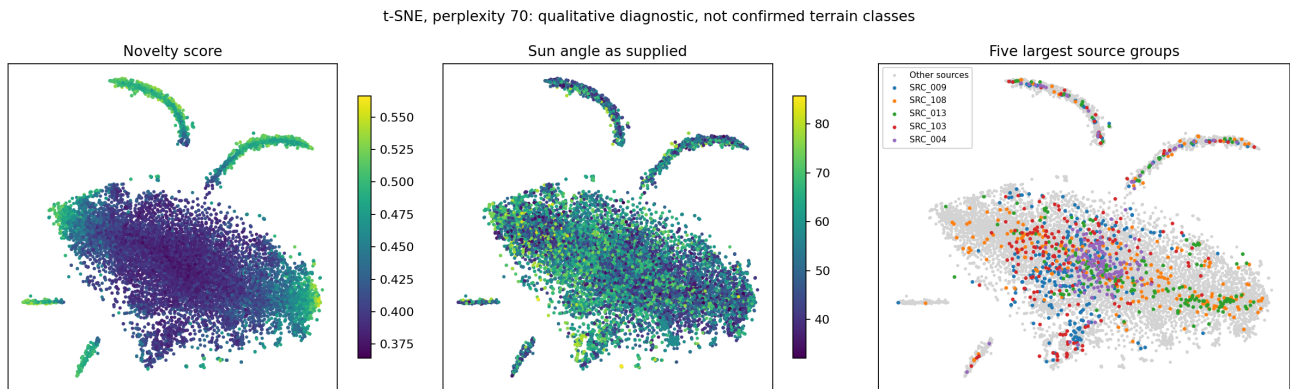


Figure 6. t-SNE at perplexity 70; novelty, supplied sun angle and five large source groups. PCA and standardization are for plotting only. The forest uses the full original latent.

The v3 latent projection is retained from the first-stage analysis. Two perplexities show broad structure, but these branches are not labelled terrain classes. Source and acquisition effects remain plausible; the between-source variance fraction is about 0.232.

Phase 2.3 Metadata and candidate context

I join only through the supplied crop-to-source index. Latitude has 24 distinct values, longitude only 0 and 180. Resolution units and sun-angle semantics are unspecified. These fields do not justify precise localization, physical dimensions or causal claims.

season	Crops	Flags	Crop rate	Equal-source rate
N-autumn	2181	6	0.275%	0.292%
N-spring	4190	2	0.048%	0.364%
N-summer	3030	8	0.264%	0.581%
N-winter	1021	1	0.098%	0.073%

resolution	Crops	Flags	Crop rate	Equal-source rate
0.25	5326	3	0.056%	0.215%
0.5	4959	13	0.262%	0.673%
1.0	137	1	0.730%	0.730%

The retained v3 flags 17 crops from eight sources. Its five selected examples all exceed the 99.7th brightness percentile, and three share SRC_154. For v4, median selected brightness falls to the 47.51st percentile and the largest source contributes 12.5% of flags, but the flagged median zero fraction is 13.36%. Source concentration and brightness improvement therefore do not remove acquisition concerns.

The primary score uses image latents only. The separately calibrated metadata experiment appears in Phase 2.4. Equal-source summaries reduce domination by large observations, but remain descriptive.

Phase 3 Candidate 1

sample_07899.jpg | SRC_128 | validation

Novelty 0.566127 exceeds 0.542471. Conditional threshold-bootstrap flag frequency is 100.0%, not a posterior anomaly probability.

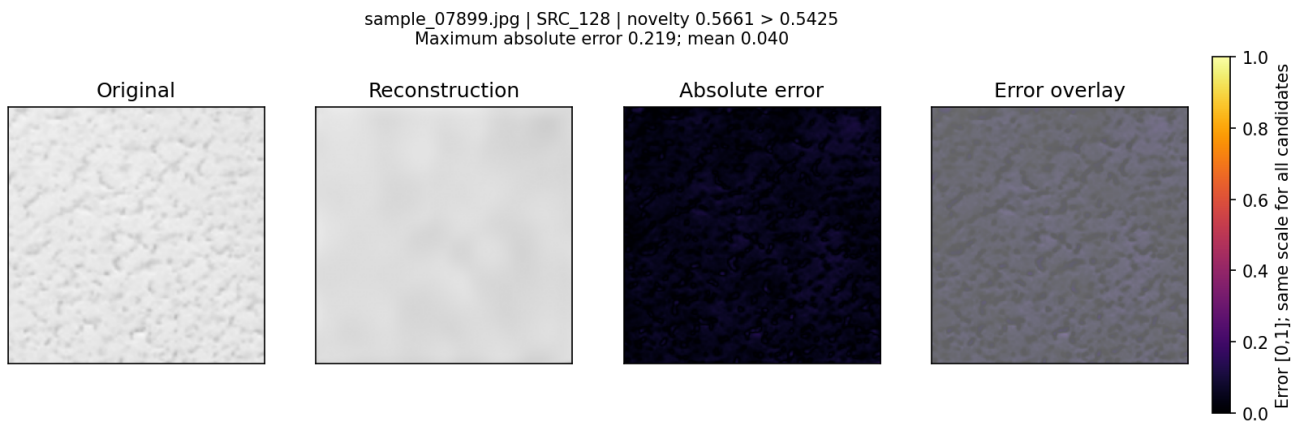


Figure 7. Input, reconstruction, absolute error and discrepancy overlay. Standardized runs additionally show the raw crop. Error refers to the model-input domain and is not direct attribution for Isolation Forest.

What I can see. A very bright crop with irregular, fine dark mottling across most of the image. The reconstruction retains broad brightness but loses the small-scale pattern. Error is distributed through that texture rather than concentrated on a single foreign object.

Possible physical explanation. A bright surface mantle, potentially seasonal frost over textured polar terrain, is a plausible ordinary explanation. The supplied latitude of 90 and N-spring category are consistent with a polar-seasonal interpretation, if those coarse metadata values are representative.

Alternative and limit. Bright dust, exposure or contrast processing could also produce this appearance. A grayscale crop cannot establish ice composition, exact location or artificial insertion. The seasonal interpretation is a hypothesis, not a catalog identification.

Supplied metadata: latitude 90.0, longitude 0.0, sun angle 56.634817, season N-spring, resolution 0.5.

Process background, not crop identification: https://www.uahirise.org/PSP_007712_2670.

Phase 3 Candidate 2

sample_08288.jpg | SRC_154 | train

Novelty 0.557379 exceeds 0.542471. Conditional threshold-bootstrap flag frequency is 100.0%, not a posterior anomaly probability.

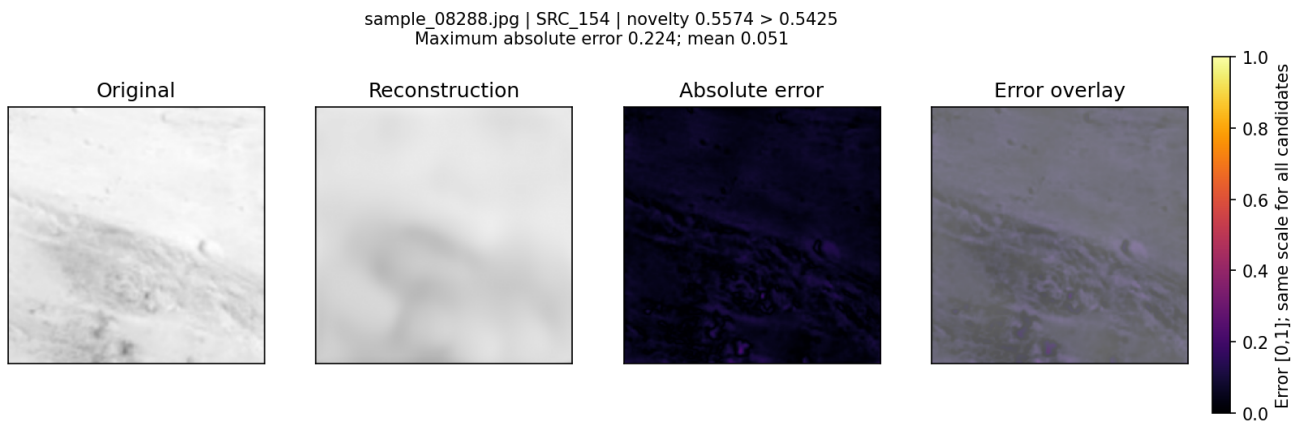


Figure 8. Input, reconstruction, absolute error and discrepancy overlay. Standardized runs additionally show the raw crop. Error refers to the model-input domain and is not direct attribution for Isolation Forest.

What I can see. A pale smooth upper region meets a darker, rougher diagonal band in the lower half. Small rounded marks and local texture remain visible. Reconstruction smooths the band, and the larger errors follow the rougher lower region.

Possible physical explanation. A contact between a fine bright surface covering and rougher exposed terrain could explain the contrast. Wind redistribution of bright dust is one plausible surface process, without requiring unusual material.

Alternative and limit. Illumination across relief or image contrast processing could produce the same contrast. The isolated crop provides no reliable slope direction, physical scale or compositional evidence. Its high latent novelty does not establish a geological anomaly.

Supplied metadata: latitude 0.0, longitude 180.0, sun angle 54.770875, season N-summer, resolution 0.5.

Process background, not crop identification: <https://www.jpl.nasa.gov/images/pia03789-cerberus-wind-streaks/>.

Phase 3 Candidate 3

sample_04609.jpg | SRC_131 | train

Novelty 0.556367 exceeds 0.542471. Conditional threshold-bootstrap flag frequency is 100.0%, not a posterior anomaly probability.

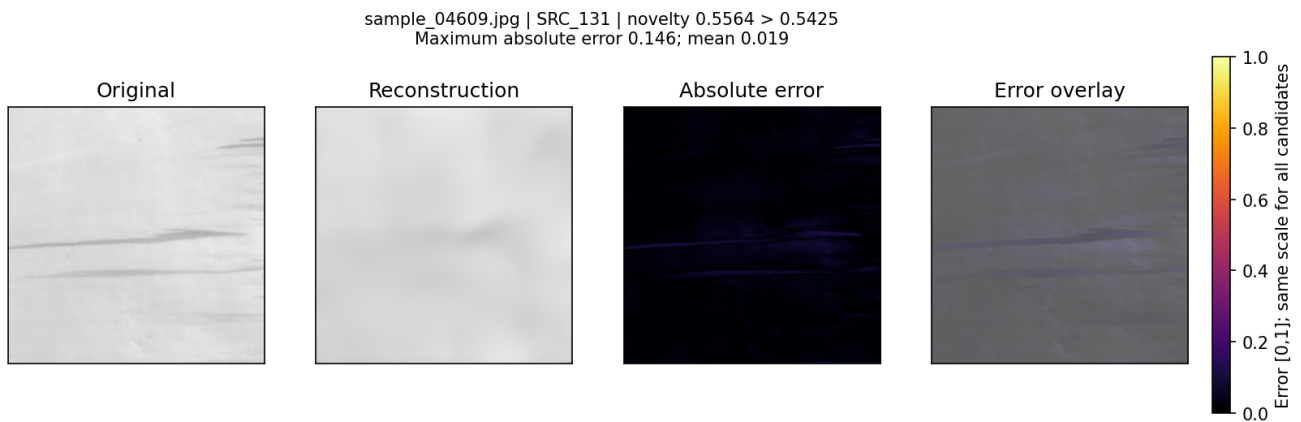


Figure 9. Input, reconstruction, absolute error and discrepancy overlay. Standardized runs additionally show the raw crop. Error refers to the model-input domain and is not direct attribution for Isolation Forest.

What I can see. Several thin, nearly parallel dark streaks cross a bright, relatively smooth background; two long streaks dominate the middle. The decoder blurs the narrow lines, so the error map traces them even though average reconstruction error is low.

Possible physical explanation. Wind-related removal or redistribution of a bright dust coating could expose darker streaks. This is an ordinary Martian process that can create strong directional contrast.

Alternative and limit. Shadows, image striping or another acquisition artifact remain possible. Orientation alone cannot distinguish a wind streak from these alternatives, and neither wind direction nor slope can be recovered reliably from this crop.

Supplied metadata: latitude 10.0, longitude 180.0, sun angle 60.540464, season N-autumn, resolution 0.5.

Process background, not crop identification: <https://www.jpl.nasa.gov/images/pia07132-wind-streaks-among-flows/>.

Phase 3 Candidate 4

sample_00840.jpg | SRC_154 | train

Novelty 0.555589 exceeds 0.542471. Conditional threshold-bootstrap flag frequency is 100.0%, not a posterior anomaly probability.

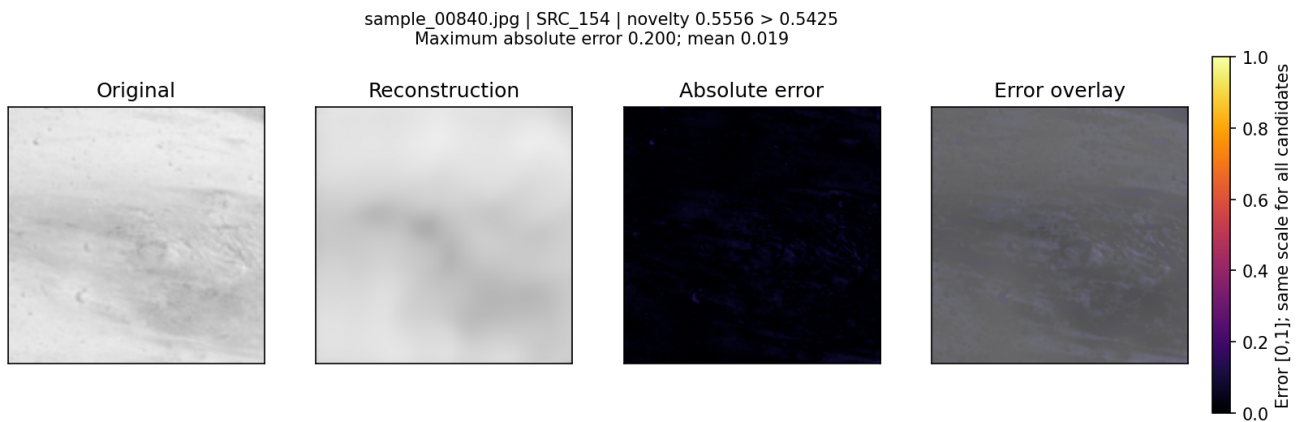


Figure 10. Input, reconstruction, absolute error and discrepancy overlay. Standardized runs additionally show the raw crop. Error refers to the model-input domain and is not direct attribution for Isolation Forest.

What I can see. A bright background contains a broad diagonal textured band and scattered small rounded depressions or marks. Reconstruction preserves the diffuse band but removes many small details. Error follows the textured band and local marks.

Possible physical explanation. Uneven bright dust cover over a rougher surface could produce this combination of smooth and textured patches. The marks may include small depressions, but their origin cannot be resolved here.

Alternative and limit. Shading or contrast processing could mimic material differences. This crop shares source SRC_154 with two other selected examples, so their similarity may reflect one acquisition rather than independent unusual geology.

Supplied metadata: latitude 0.0, longitude 180.0, sun angle 54.770875, season N-summer, resolution 0.5.

Process background, not crop identification: <https://www.jpl.nasa.gov/images/pia03789-cerberus-wind-streaks/>.

Phase 3 Candidate 5

sample_05391.jpg | SRC_154 | train

Novelty 0.555249 exceeds 0.542471. Conditional threshold-bootstrap flag frequency is 100.0%, not a posterior anomaly probability.

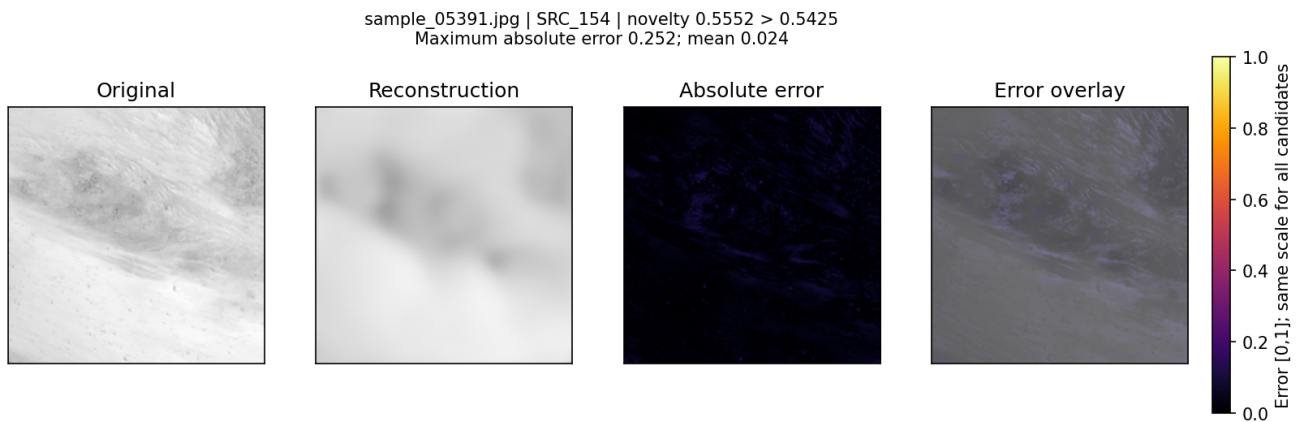


Figure 11. Input, reconstruction, absolute error and discrepancy overlay. Standardized runs additionally show the raw crop. Error refers to the model-input domain and is not direct attribution for Isolation Forest.

What I can see. A rough, striated upper region borders a brighter, smoother lower area along an oblique boundary. Fine linear textures dominate the upper part, and the reconstruction reduces them to broad smooth forms. The error map mainly follows that lost texture.

Possible physical explanation. A transition between a dust-mantled surface and a more exposed textured substrate could explain the boundary. Wind erosion or deposition is a plausible process, but this image alone does not identify the landform.

Alternative and limit. Relief-dependent lighting or radiometric processing remains an alternative. Three of the five selected crops come from SRC_154; source conditions may influence the repeated selection. No injected object is confirmed by this panel.

Supplied metadata: latitude 0.0, longitude 180.0, sun angle 54.770875, season N-summer, resolution 0.5.

Process background, not crop identification: <https://www.jpl.nasa.gov/images/pia03789-cerberus-wind-streaks/>.

Phase 4 My engineering notes

v1: Baseline

I trained a 128-dimensional MSE autoencoder from scratch. Validation MSE was 0.002694; fine texture remained blurred.

v2 and v3: Loss and capacity

Adding SSIM raised validation SSIM from 0.6611 to 0.6717 with a small MSE cost. Increasing the vector to 256 gave MSE 0.002599 and SSIM 0.6738. More parameters improved reconstruction modestly, not verified anomaly accuracy.

Threshold and repetition

The initial skew-adjusted fence exceeded every score. I retained it, inspected the multimodal distribution and adopted a fixed mixture envelope. Increasing the forest to 2,000 trees improved repeatability. Independent v3 runs still had flag Jaccards 0.150 and 0.294.

v4 and v5: Contrast and footprint

Normalization reduced a fixed illumination response, but selected borders became prominent. Excluding connected exact zeros left near-black fragments and weakened forest agreement. I did not promote these variants.

v6: Near-black border fill

I tested a fixed near-black mask and nearest-valid-pixel fill. Border response decreased, but defect response and forest stability failed the predefined criteria. Filled regions introduced stretched stripes. This was a completed negative experiment, not an abandoned variance mask.

v7: Gradient loss

I added 0.1 times adjacent-pixel gradient error to 0.9 MSE + 0.1 (1-SSIM). MSE improved 1.68% and gradient error 0.52%. Four of five selected crops were still extremely bright; the darker crop also contains borders. The brightness confound was not solved. The protocol's 0.8 MSE description differed from the actual 0.9 implementation; the saved configuration and source define the executed experiment.

v8: Geometric augmentation and optional fusion

I added batch flips and quarter-turn rotations to the v7 objective. Validation reconstruction worsened and image-only screening produced zero flags. Lower effective rank does not prove a better representation. A separate post-encoding metadata experiment is reported with its own threshold; zero flags are retained without manufacturing five examples.

Final decision

I keep v3 as the reference, disclose its brightness and independent-run limitations, and retain all fourteen completed runs as evidence. No hidden-label accuracy is claimed.

References and reproducibility

Competition inputs. DA PS 26'.docx, DATA_DESCRIPTION.docx, RULES AND REGULATIONS.docx and the supplied archive.

SSIM. <https://www.cns.nyu.edu/~lcv/ssim/>

Resize-convolution. <https://distill.pub/2016/deconv-checkerboard/>

Isolation Forest. <https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.IsolationForest.html>

Gaussian mixtures. https://scikit-learn.org/stable/auto_examples/mixture/plot_gmm_selection.html

Connected-component labelling.

<https://docs.scipy.org/doc/scipy/reference/generated/scipy.ndimage.label.html>

The readable Colab notebook contains the preprocessing, model, training and scoring code directly. Completed configurations, checkpoints, manifests, per-epoch histories and numerical tables accompany the analysis. A fresh runtime needs the supplied private dataset ZIP. No pretrained weights or outside training images are used.

The recorded execution was local. Epoch wall times include interruptions and are not controlled hardware benchmarks. The independent checks are a limited sample of seeds and source assignments. Hidden labels would be needed to establish actual detection performance.

Private repository access, organizer collaborators and team details remain competition logistics. Local packaging does not submit the project.